

A novel dynamic survivable routing in WDM optical networks with/without sparse wavelength conversion

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Abstract

In this paper, we study the dynamic survivable routing problem, both in optical networks without wavelength conversion and in optical networks with sparse wavelength conversion, and propose a novel hybrid algorithm for it based on the combination of mobile agents technique and genetic algorithms (GA). By keeping a suitable number of mobile agents in the network to cooperatively explore the network states and continuously report cycles (that are formed by two disjoint-link routes) into the routing tables, our new hybrid algorithm can promptly determine the first population of cycles for a new request based on the routing table of its source node, without the time consuming process associated with current GA-based lightpath protection schemes. We further improve the performance of our algorithm by introducing a more advanced fitness function that is suitable for both the above networks. Extensive simulation studies on the ns-2 network simulator show that our hybrid algorithm achieves a significantly lower blocking probability than the conventional survivable routing algorithms for all the cases we studied.

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1. Introduction

In recent years, optical networks using Wavelength Division Multiplexing (WDM) technology have been generally regarded as the most promising technology to form the backbone of the next-generation Internet [1]. A WDM optical network consists of a set of *wavelength*

cross-connects (WXC)s interconnected by fibre links in an arbitrary topology. In all-optical WDM networks, end-users communicate with each other via all-optical channels that are referred to as *lightpaths*.

There are two kinds of WXC:s: *wavelength selective cross-connect* (WSXC) and *wavelength interchange cross-connect* (WIXC). A WSXC can switch an optical signal from an input fibre to an output fibre on the same wavelength without performing optoelectronic conversion. WDM networks that use WSXC:s are referred to as *wavelength selective networks*. In a wavelength selective network, a lightpath requires the

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same wavelength on all the links along its route, which is referred to as the *wavelength continuity constraint* [2]. Unlike a WSXC, a WIXC can switch an optical signal from an input fibre to an output fibre on a different wavelength by using *wavelength converters*; thus the wavelength continuity constraint is disposed. WDM networks that use WIXCs are referred to as *wavelength convertible networks*. Wavelength converters are expensive devices, so should only be used at a few strategic nodes. Such a network is called one with *sparse wavelength conversion*. A node is capable of *full-range* wavelength conversion if a wavelength channel on any input of the node can be converted to any wavelength channel on any output; a node is capable of *limited-range* wavelength conversion if a wavelength channel on any input of the node can only be converted to particular wavelength channels on any output [3].

1.1. Routing and wavelength assignment

The problem of setting up lightpaths by routing and assigning a wavelength to each connection such that no two lightpaths use the same wavelength on the same link is called the *routing and wavelength assignment* (RWA) problem. The traffic model assumed by an RWA algorithm can be either *static* or *dynamic*. In a static traffic model, a set of connection requests is given in advance. The objective is typically to minimize the number of wavelengths, converters, or other cost parameters. This problem can be formulated as a mixed-integer linear program, which is NP-complete [2].

In a dynamic traffic model, the connection requests arrive and depart one by one in a random manner and the setting up lightpaths depends on the current state of the network, thus the problem is more difficult and challenging than in the static traffic model. Here, the objective is to increase the acceptance ratio (or equivalently, to decrease the blocking probability) of the connections. This problem is usually solved by heuristic algorithms. There are three major approaches to this problem: the *fixed*, *fixed-alternate*, and *dynamic* approach [2]. The first two approaches use a set of pre-computed shortest paths to establish a lightpath. These approaches are simple, have small setup time, and low control overhead, but usually result in high blocking probability. The third approach, i.e. dynamic, is more efficient than the first two in terms of blocking probability, because the route is chosen based on the current state of the network. However, dynamic routing usually requires a long setup time and a high control overhead.

1.2. Lightpath protection

WDM networks are prone to component failures, which may seriously damage end-user applications. Therefore, it is imperative to have networks that are capable of preventing such failures; such networks are referred to as *survivable networks* [4]. In survivable WDM networks, the lightpath that carries traffic during normal operations is known as the *primary lightpath*. When a primary lightpath fails, the traffic is rerouted over a new lightpath known as the *backup lightpath*. The network can be protected against failures by *protection* and/or *restoration* schemes. In protection schemes, both the primary and backup lightpaths are computed before a failure occurs. In restoration schemes, a backup lightpath is discovered dynamically for a primary lightpath after a failure occurs. Protection schemes have faster recovery times and provide a 100% recovery guarantee, but require more network resources than restoration schemes.

Lightpath protection schemes require the finding of a pair of diverse routes that are disjoint links and form a *cycle* (a primary path and a backup path) for each connection request. This problem is also known as *survivable routing* (i.e. diverse routing). Lightpath protection schemes can be divided into *dedicated* and *shared* protection. In dedicated protection, the backup resources are used for at most one lightpath. In shared protection, the backup resources are reserved for multiple connections as long as their primary lightpaths do not fail simultaneously. It has been shown that shared protection schemes are more efficient than dedicated protection schemes; therefore they are widely used in survivable WDM networks.

A network failure can be caused by either a link failure or a node failure. However, most modern node devices are equipped with built-in redundancy to improve their reliability. Thus, current research mainly focuses on the *single link failure* model [2], in which at most one link can fail at any instance of time (failures do not accumulate and when a link fails, any link that has failed earlier has been repaired).

1.3. Related work

Mohan et al. [5] proposed two algorithms, called the primary dependent backup wavelength assignment (PDBWA) and the primary independent backup wavelength assignment, (PIBWA) for the problem of dynamic survivable routing in WDM optical networks. These algorithms are very attractive in that they use a backup multiplexing technique, which allows backup

lightpaths to use the same wavelength on the same link if their primary lightpaths are disjoint links, and hence, efficiently utilize network resources and significantly improve network performance [5]. Different from the PIBWA, the PDBWA requires the same wavelength on the primary and backup lightpaths for a request; and the PIBWA selects a pair of primary and backup lightpaths such that its total cost is lowest, while the PDBWA selects such a pair of lightpaths that the primary lightpath has the lowest cost. The authors claimed that the PIBWA has the advantage of better network performance in terms of blocking probability, compared with the PDBWA. However, if we modified the PDBWA to allow use of different wavelengths on the primary lightpath and the backup lightpaths of a request as the same as PIBWA (hereafter called PDBWA-New), the PDBWA-New can achieve better network performance in terms of blocking probability in comparison with the PIBWA. This is described by simulations in Section 4.

In addition, a property of those algorithms is that they use the fixed-alternate routing scheme to reduce complexity. As a consequence, they may limit network performance in terms of blocking probability; therefore their performance could be improved by adopting a dynamic routing approach, in which the lightpaths are established adaptively-based on the current network state. However, such a problem has been shown to be NP-complete [4,6], so it would be preferable to solve it by a heuristic approach within a reasonable computation time. In [7], Bisbal et al. inherited the PIBWA algorithm and proposed a dynamic routing heuristic using a genetic algorithm, namely the fault-tolerance GA-based RWA (FT-GRWA) algorithm. By using a GA approach, the FT-GRWA algorithm can provide much better performance than the PIBWA algorithm with a reasonable computation time, but it still has some drawbacks. It requires the time consuming process of random searching for the first population of cycles whenever a new connection request arrives, and this can result in a significantly large setup delay. Furthermore, the authors defined the cost function as the sum of the cost of the primary path and the cost of the backup path, i.e. the cost of a unit of the network resource used for a primary lightpath and for a backup lightpath is the same. Thus, a cycle with higher primary path cost could be selected if the cost of its backup path is small enough to create a smaller total cost. This could result in a higher blocking probability because a higher primary path cost means more resources are reserved.

However, the wavelength continuity constraint usually results in a high blocking probability in a wavelength-routed WDM network, and a common

approach to alleviating this constraint is the adoption of wavelength converters. In addition, previous studies showed that conventional RWA algorithms may not work well in an environment with sparse or/and full wavelength conversion [3]; the above algorithms not only did not consider the presence of wavelength converters on the networks but also cannot be applied directly to WDM networks with wavelength conversion capability. Thus, in a WDM network with sparse wavelength conversion, the dynamic survivable routing problem needs deliberate study to take full advantage of the gain from wavelength conversion [8,9]. With such an observation, we proposed an RWA algorithm based on the combination of mobile agents technique and GA [10]. However, that paper only considered the normal nonsurvivable routing problem; the fitness function in the algorithm may be not good because it has to compute two values for each route with wavelength converters, one for the case with wavelength conversion and another for the case without. Hybrid GA/heuristic algorithms were also proposed in [11]. Although these algorithms showed promising results compared to the various RWA algorithms, they considered the RWA a static problem and did not consider failures of the networks.

1.4. This work

In this paper, we investigate the dynamic survivable routing problem, both for optical networks without wavelength conversion and for optical networks with sparse wavelength conversion, using a shared backup scheme and different wavelengths for primary and backup lightpaths, as described in [5]. To overcome the above mentioned drawbacks of the FT-GRWA method, we propose a novel hybrid algorithm based on a combination of ant-based mobile agents [12] and genetic algorithms [13]. By keeping a suitable number of mobile agents in the network to explore the network states cooperatively and continuously report the cycles into the routing tables, we can promptly determine the first population of cycles for a new connection request based on the routing table of its source node, without requiring a time consuming random search of first population cycles as the case of FT-GRWA method. Furthermore, using mobile agent can determine a better first population than that generated by random searching, thus the probability with which the evolution process will occur is decreased. We also propose a new fitness function that not only utilizes the network resources more efficiently for establishing a protected lightpath in both types of

optical networks mentioned above, but also enhances that presented in [10]. In addition, we introduce a general formula for determining the key parameter in the new fitness function. Our algorithm is very attractive in that it provides low blocking probability by adopting the shared protection scheme, while it has a small computation time through the combination of GAs and mobile agent searching techniques.

1.5. Organization of this paper

The rest of this paper is organized as follows: in Section 2, we briefly review the background of GA and mobile agent techniques for optical network routing. Section 3 presents the new hybrid algorithm. Section 4 provides the simulation results and discussion. Finally, we conclude the paper in Section 5.

2. Background

In this section, we briefly introduce the main principles of genetic algorithms and mobile agent techniques for routing and wavelength assignment.

2.1. Genetic algorithms

Genetic Algorithms (GA) are a class of probabilistic searching algorithms based on the mechanism of biological evolution. A GA begins with an initial population of individuals; each of them represents a feasible solution to the problem being tackled. Then the GA applies a set of genetic operations, such as crossover or mutation, to the current population to generate a better one. This process is repeated until a good solution is found or until a predefined number of iterations is reached [13].

2.2. Ant-based mobile agents routing

Routing in communication networks can be resolved efficiently by means of Ant Colony Optimization (ACO) [12,14], in which the routing solution is built using mobile agents that simulate the foraging behaviour of ants in their foraging of network states. These collective agents indirectly communicate through pheromone trailing (stigmergy) in the environment, and an agent can find a ‘good’ route in terms of shortest, least congested path from the source to the destination by following the pheromone trails of others.

Recently, Ngo et al. [14] proposed an ant-based mobile agents algorithm for solving the dynamic RWA problem. In this algorithm, a network node is equipped with a probabilistic pheromone table that

contains the selection probabilities of neighbour nodes when an ant is moving towards its destination. These probabilities serve for selecting a neighbour as the next hop when finding a route for a connection request. Ants are launched from each node to a randomly selected destination at regular time intervals. Each ant is considered to be a mobile agent: it collects information on its trip, performs pheromone table updating on visited nodes and continues to move forward. Whenever an ant visits a node, it updates the pheromone table elements with the information gathered during its trip. Using the behaviour of natural ants as shown in Ant-Colony Optimization principles [12], this process ensures that the network congestion state will be well reflected in the routing tables. Thus, when a new connection request arrives at its source node, its routes can be determined promptly from the routing tables: starting from the source node, the next hop will always be the neighbour that has the highest selection probability, and this principle is applied until reaching at the destination node.

With the above method, the routes are already determined upon the arrival of a connection request, thus Ngo’s algorithm is an attractive tool in GA-based dynamic survivable routing algorithms for generating the first population of cycles.

3. Hybrid algorithm based on a combination of GA and mobile agent technique

In dynamic lightpath protection, it is critical to determine the solution within a short computation time. We observe that GA-based approaches may incur a very large setup delay because they require a time consuming random search process to generate the first population of cycles after the arrival of a new request. On the other hand, mobile agent based approaches can assign the routes promptly when a connection request arrives. These observations motivated us to propose in the present paper a new hybrid algorithm for the dynamic survivable routing that first uses the mobile agents technique to find P_{size} cycles for the first population, and then uses a GA to find the best cycle for setting up the lightpath. The hybrid algorithm is executed at the arrival time of a connection.

3.1. Hybrid algorithm

In our algorithm, we use the presentation of individuals, genetic operators, and reproduction process in the same way as described in [7]. An individual is presented as a cycle that is formed from two disjoint-link routes. Each route is encoded by integer numbers,

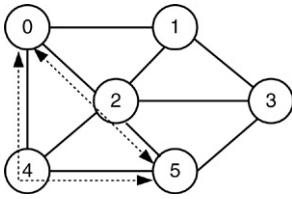


Fig. 1. Two disjoint-link routes form a cycle: $(0\ 2\ 5)(0\ 4\ 5) \Rightarrow (0\ 2\ 5)(5\ 4\ 0) \Rightarrow (0\ 2\ 5\ 4\ 0)$.

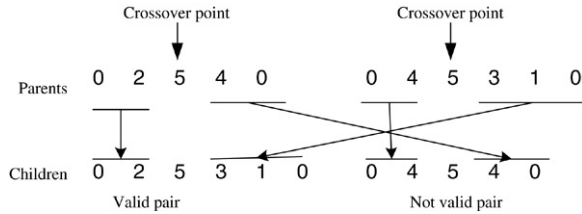


Fig. 2. An example of crossover operation.

each of which identifies a node of the route. For illustration, Fig. 1 shows an example of a network topology and a cycle between node 0 and node 5: the coding of two routes from node 0 to node 5 are $(0, 2, 5)$ and $(0, 4, 5)$ that form the cycle $(0, 2, 5, 4, 0)$.

The initial population consists of P_{size} cycles that are generated by the ant-based mobile agents technique (where P_{size} is a design parameter). This population is then evolved by genetic operators: the crossover and mutation operators. The crossover operator is applied to a pair of cycles that has at least one node in common. The children are generated by interchanging the second half of their parents, as illustrated in Fig. 2. The children cycles must have two halves that are disjoint links.

In the mutation operator, a node, say m , from a cycle is randomly selected. The route portion from the source node to node m remains intact and the route portion from node m to the source is created again. This newly created route portion must traverse the destination node in case node m is located before the destination node in the original cycle. Note that in the next cycle has to satisfy the disjoint links condition.

After applying the genetic operators above, the reproduction stage selects P_{size} fittest individuals that have the highest fitness value from both parents and children, for the next generation. This process is repeated until the stopping condition is fulfilled and the best individual is selected.

Let G denote the maximum number of generations and S denote the satisfactory cost value of the primary route between a node-pair with its initial value being the cost value of the shortest route between the node-pair.

The pseudo code of the GA part of our hybrid algorithm can be summarized as follows:

```

 $t_G \leftarrow 0$ ;
Evaluate fitness values for individuals of the first
population;
 $S \leftarrow$  shortest distance between  $(s, d)$  nodes;
while ( $t_G < G$  AND not exist a cycle in which the
length of the primary route is lower or equal  $S$ ) do
  Crossover & evaluate fitness value for children;
  Mutation & evaluate fitness value for children;
  Select  $P$  fittest;
   $S \leftarrow S + 1$ ;
   $t_G \leftarrow t_G + 1$ ;
end while
Select the best cycle;

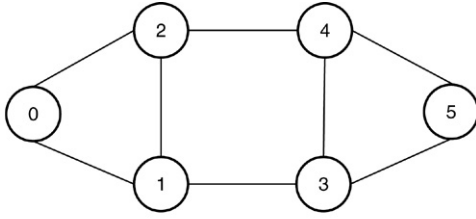
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It is of interest to note that the proposed algorithm is centralized, i.e. there is a control node in charge of receiving requests, determining routes and assigning wavelengths. The movement of the mobile agents (and their activity for building the tables) is emulated in the control node of the network. When a request arrives at the control node, the cycles that have been obtained by emulating the movement of mobile agents form the first population, and then the availability of wavelengths and the length of the primary lightpath are checked. If no wavelength is available or the primary lightpath is not the shortest route between the node-pair, the population is evolved. After the evolution process, if no cycle has been found with an available wavelength, the request is blocked.

3.2. Mobile agents technique for finding P_{size} cycles

In this section we present the mobile agents technique to find P_{size} cycles as a first population for the hybrid algorithm. This technique is based on an ant-based agent approach for dynamic RWA in WDM networks [14].

To ensure that a first population of P_{size} cycles is available for a node-pair, each node is equipped with a new cycles routing table and a pheromone table $R_i = [r_{n,d}^i]_{k_i, N-1}$ that has $N - 1$ rows (N is number of network nodes) and k_i columns. Each row corresponds to a destination node and each column corresponds to a neighbour node; the value $r_{n,d}^i$ is used as the selection probability of neighbour node



Destinations	Neighbor		
	1	4	5
0	$r_{1,0}^3=0.6$	$r_{4,0}^3=0.3$	$r_{5,0}^3=0.1$
1	$r_{1,1}^3=0.8$	$r_{4,1}^3=0.2$	$r_{5,1}^3=0.0$
....			
5	$r_{1,5}^3=0.0$	$r_{4,5}^3=0.2$	$r_{5,5}^3=0.8$

Fig. 3. A simple network and its pheromone table of node 3.

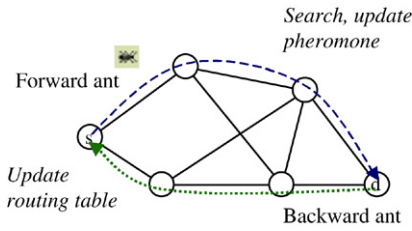


Fig. 4. Ant movements and updating task.

n when an ant is moving toward its destination node d . An example of a pheromone table is illustrated in Fig. 3:

The cycles routing table contains $N - 1$ entries, and each entry contains a list of P_{size} cycles to a destination node; these P_{size} cycles serve as the first population for a future request between the current node and the destination node. The role of mobile agents is two-fold: they must continuously update both pheromone and cycles routing tables. Whenever an ant visits a node, as illustrated in Fig. 4, it updates the routing table (pheromone updating). Suppose an ant moves from the source node s to destination node d following the route $(s, \dots, i - 1, i, \dots, d)$, it will update the entry in the routing table of node i , which corresponds to the source node s , in a manner so that the probability of selecting neighbour $i - 1$ is increased and the probabilities of selecting other neighbours are decreased. Namely, assuming an ant visits node i at time t , then the values for routing entry in time $t + 1$ are determined by the following formula:

$$r_{i-1,s}^i(t+1) = \frac{r_{i-1,s}^i(t) + \delta r}{1 + \delta r}; \quad i > 0$$

$$r_{n,s}^i(t+1) = \frac{r_{n,s}^i(t)}{1 + \delta r}; \quad n \neq i - 1, i > 0.$$

Here, δr is the reinforcement parameter and is derived from the data collected by the ant.

$$\delta r = \frac{\theta}{\delta l} + (1 - \theta) \cdot \delta w; \quad 0 \leq \theta \leq 1$$

where δl corresponds to the length of the path in terms of number of hops, δw corresponds to the percentage of free wavelengths of the path that ant moved along, and θ is a scalar parameter that can be used to adjust the emphasis of path length versus free wavelength percentage. For the normalization of pheromone updating, these two factors are computed as follows:

$$\delta l = \beta \cdot (e^{\frac{-1.0}{l}} - e^{-1}), \quad \delta w = e^{\gamma \cdot w} - 1$$

where l is the number of hops of the path, w is the percentage of free wavelengths of the path and w is deducted from the corresponding wavelength mask in the ant stack. Here θ , β and γ are designed parameters that can be adjusted in the experiment.

Ants are released from each node to a randomly selected destination every t_{ant} time unit (where t_{ant} is a design parameter). When an ant moves from a source to a destination, its next hop is determined stochastically: a neighbour is selected according to its selection probabilities in the pheromone table [14].

Whenever an agent reaches its destination, it returns to its source to explore a cycle. The rule for backward moving is similar to that for forward moving. In addition, the agent checks so as not to move onto a visited link, hence when it gets to its source, all the visited nodes will form a cycle between the source-destination nodes. At this time, it updates the cycles routing table of its source node and then it is killed. Note that an ant is also killed if on the path it revisits a node which makes the path containing a cycle. If this table contains fewer than P_{size} cycles in the cycle list corresponding to its source node, the new cycle is directly inserted into that list. Otherwise (i.e. the list already contains P_{size} cycles), the new one will replace an old cycle in the list based on a *First-In-First-Out* policy. This replacement is necessary to ensure randomness in the first population. The pseudo code of this process can be described as follows:

```

If the new route is different from any available routes
  in the route list of source node Then
  If number of routes in the route list less than  $P$  Then
    Insert agent's new route into the list
  Else
    Replace an existing route with the agent's new
    route based on the First-In-First-Out policy
  End if
End if

```

Since we always keep a suitable number of mobile agents in the network to explore the network states cooperatively and continuously update the pheromone tables and cycle routing ones, each cycle routing table will contain P_{size} cycles for each destination after an initialization period; those cycles are always updated based on the current network state. The updating rules take into account both the free wavelength of links and the path length. Since the network is dynamic, ants will traverse different cycles, not only the shortest path. By consequence, the cycle routing tables will change according to the network state. Thus, they can serve as the first population of P_{size} cycles upon the arrival of a request.

The number of mobile agents is a trade-off parameter of the hybrid algorithm. The larger the number of mobile agents searching the network, the more frequently the cycle routing tables are updated. We determine a suitable number of mobile agents by extensive simulations in Section 4. We first give it a small initial value and then increase this value until the hybrid algorithm obtains a good network performance.

3.3. New fitness function

To yield the best performance for dynamic survivable routing, the key idea is to enable the selection of the cycle in which the primary lightpath is the shortest available path and the backup lightpath uses a minimum of free channels. Furthermore, to take the advantage of the wavelength conversion capability, a primary route with wavelength conversion should only be chosen if no route without wavelength conversion is available, whereas a backup route should exploit the wavelength conversion to reduce the number of free channels it requires. This is because we can use wavelength converters to convert the optical signal from a free channel to one which is being used by another backup lightpath.

In the following we propose a new fitness function which takes into account all the above ideas.

3.3.1. The case of optical networks without wavelength conversion

The cost of a cycle will be computed from the cost of its primary lightpath and its backup lightpath. The fitness function is defined as the inverse of the cost of the cycle.

Let CP be the cost of the primary lightpath. CP is defined as the number of hops, assuming there is at least one available wavelength on the primary path. If several wavelengths are available, the lowest indexed among them is assigned to the lightpath. If there is no wavelength available, CP is infinite.

Let CB be the cost of a backup lightpath and λ -channel denote a wavelength on a fibre. Given a fibre f , let $c_{f,w}$ ($w = 0, \dots, W$) denote each λ -channel on that fibre (where W is total number of wavelengths on a fibre); $c_{f,w}$ is 1 if its λ -channel is used neither by any primary lightpath nor by any backup lightpath, 0 if its λ -channel is used by a set of backup lightpath Φ and its primary route is disjoint links with the primary route of each backup lightpath in the set Φ , and infinite otherwise. Then, the cost of the backup lightpath for each wavelength w , denoted by CB_w , is computed as the sum of the costs of each λ -channel of the route.

$$CB_w = \sum_{f \in \text{route}} c_{f,w}. \quad (1)$$

The cost of the backup lightpath is taken as the minimum over CB_w and this wavelength is assigned to the backup lightpath.

$$CB = \min_{w=0,\dots,W} \{CB_w\}. \quad (2)$$

A cycle ($s - d - s$) is interpreted as two $s - d$ routes, one for the primary lightpath and the other for the backup lightpath. One way to do that is to let the first portion of the cycle represent the route of the primary lightpath and assign the second portion to the backup lightpath. The cycle could be also interpreted inversely, that is, its first portion is assigned to the backup lightpath, and the second portion to the primary one. The cost of the cycle is computed assuming both interpretations and the one with the lower cost is chosen. For each interpretation, Bisbal [7] defined the cost of the cycle as:

$$C = CP + CB + \left(\frac{1}{N}\right) \cdot h \quad (3)$$

where N is the number of network nodes and h is the number of hops of the primary lightpath.

Since in Bisbal's definition the cost of a free channel on a link of a primary lightpath or a backup lightpath

is the same when evaluating the cost of a cycle, it is possible that a cycle with a higher primary lightpath cost could be selected if the cost of its backup lightpath is small enough to give a smaller total cost. The following example illustrates this situation.

Consider the NSF topology in Fig. 6(a) with two wavelengths per link. A primary lightpath (0, 1, 7) is established between nodes 0 and 7, and a backup lightpath (0, 3, 4, 6, 7) is established between the same nodes. The wavelength λ_0 is assigned to both the primary and backup lightpaths. Assume now there is a request for the establishment of a protected connection from node 6 to node 11. We then need to compute the cost of the best cycle (6, 4, 3, 11, 12, 10, 7, 6), which represents two disjoint-link routes: (6, 4, 3, 11) and (6, 7, 10, 12, 11).

Case (a). The route (6, 4, 3, 11) serves as the primary lightpath. The route's cost is $CP = 3$ and it uses wavelength λ_1 . The backup lightpath (6, 7, 10, 12, 11) travels through the link (6, 7) that is shared with the backup lightpath (0, 3, 4, 6, 7). Thus, the costs of the backup lightpaths for wavelength λ_0 and λ_1 are determined as follows according to (1):

$$CB_0 = 0 + 1 + 1 + 1 = 3$$

$$CB_1 = 1 + 1 + 1 + 1 = 4.$$

Then the minimum cost of a backup lightpath is $CB = 3$ according to (2). The cycle's cost in this case is:

$$C = 3 + 3 + 3 \cdot 1/14.$$

The pair of wavelengths used for primary and backup lightpaths are λ_1 and λ_0 , respectively.

Case (b). The route (6, 7, 10, 12, 11) serves as the primary lightpath using wavelength λ_1 and its cost is $CP = 4$. The backup lightpath (6, 4, 3, 11) has two links (6, 4), (4, 3) that are shared with the backup lightpath (0, 3, 4, 6, 7). Thus, the costs of the backup lightpaths for wavelength λ_0 and λ_1 are:

$$CB_0 = 0 + 0 + 1 = 1$$

$$CB_1 = 1 + 1 + 1 = 3.$$

Then the minimum cost of a backup lightpath is $CB = 1$ according to (2). The cycle's cost in this case is:

$$C = 4 + 1 + 4 \cdot 1/14.$$

The pair of wavelengths used for primary and backup lightpaths are λ_1 and λ_0 , respectively.

In this example, according to Bisbal's definition, it is easily seen that case (b) is selected; however case

(a) should have been selected because it has the shorter primary lightpath.

Note that, if the cost of a free channel on a link of a primary or backup lightpath is seen the same, then the total numbers of used channels are 6 (3 for the primary lightpath and 3 for the backup lightpath) and 5 (4 for the primary lightpath and 1 for the backup lightpath) for case (a) and case (b) respectively, i.e. case (a) needs more network resources than case (b). However, this is not right because a backup multiplexing technique is using. As mentioned earlier, in the backup multiplexing technique, backup lightpaths can use the same wavelength on the same link if their primary lightpaths are disjoint links. This means that channels used for the backup lightpaths can be used again for different backup lightpaths of future requests. On the other hand, used channels can not be re-used by primary lightpaths. Therefore here it could not count the total numbers of channels for case (a) and case (b) being 6 and 5 respectively as above. To describe more clearly this situation, let's consider the following example:

Assume that, now there is a request for the establishment of a protected connection from node 10 to node 11. It then needs to compute the cost of the best cycle (10, 12, 11, 13, 10), which represents two disjoint-link routes: (10, 12, 11) and (10, 13, 11).

If the protected connection from node 6 to node 11 was established according to case (b), the establishment of the protected connection for this request requires two new channels for the backup lightpath and two new channels for the primary lightpath (for both cases we choose (10, 12, 11) as the primary lightpath and (10, 13, 11) as the backup lightpath and vice versa). Thus, it needs to use $2 + 4 + 2 = 8$ channels for primary lightpaths and $4 + 1 + 2 = 7$ channels for backup lightpaths for three requests.

However, if the protected connection from node 6 to node 11 was established according to case (a), the establishment of the protected connection for this request only requires two new channels for the primary lightpath (10, 13, 11) because the backup lightpath (10, 12, 11) is shared with the backup lightpath (6, 7, 10, 12, 11). Thus, it needs to use $2 + 3 + 2 = 7$ channels for primary lightpaths and $4 + 3 + 0 = 7$ channels for backup lightpaths for three requests.

This example clearly shows that a cycle with a shorter primary lightpath should have a lower cost for a request. It is also worth noting that by the example, with a similar argument as above, it supports an evidence convincing that, PDBWA with allowing different wavelengths used by primary and backup

lightpaths (i.e. PDBWA-New) will utilize network resources more efficiently than the PIBWA.

In order to ensure that the cycle with the shortest available primary path is always chosen, thus avoiding the situation illustrated in the above example, the cost of a cycle should be defined as follows:

$$C = CP + \alpha \cdot CB \quad (4)$$

where $\alpha \in [0, 1]$ is a design parameter. The α parameter should be chosen such that the cycle consisting of a shorter primary route has a smaller cost; If CP_1 , CB_1 , CP_2 , CB_2 are costs of the primary and backup lightpaths of two cycles for a connection request respectively, and assuming that $CP_1 < CP_2$ (that means $CP_2 \geq CP_1 + 1$), $\forall CB_1, CB_2$, then should meet the following requirement:

$$CP_1 + \alpha \cdot CB_1 < CP_2 + \alpha \cdot CB_2.$$

If there is an available wavelength for the backup lightpath, its minimum cost is zero (all its links are shared with other backup lightpaths) and its maximum cost is denoted by L , which is the length of the longest path. The value L is determined off-line based on the algorithm described in [15]. Thus the above inequality becomes:

$$CP_1 + \alpha \cdot L < (CP_1 + 1) + \alpha \cdot 0$$

which is equivalent to:

$$\alpha < \frac{1}{L}. \quad (5)$$

3.3.2. The case of optical networks with sparse wavelength conversion

In this case, the fitness function is defined as the inverse of the cost of a cycle, similar to Section 3.3.1 above. However, the cost evaluation of a cycle is more complex. This is because optical networks with sparse wavelength conversion are being considered, in which routes may or may not consist of wavelength converters. That is, the two following cases must be distinguished:

Case (a): A route without wavelength conversion. When a route without wavelength conversion is used by a primary lightpath or backup lightpath, the cost of the primary lightpath or the cost of the backup lightpath is computed as in Section 3.3.1.

Case (b): A route with wavelength conversion. When a route consists of t_{wc} wavelength converters, the scheme proposed in [16] is used to divide the route into $(t_{wc} + 1)$ segments $s_1, s_2, \dots, s_{t_{wc}+1}$ as illustrated in Fig. 5.

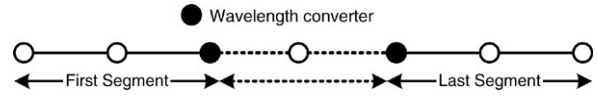


Fig. 5. A route and its segments.

When a route consists of wavelength converters and is used as a primary route, if there is an available wavelength on the route without using wavelength conversion, the cost of the primary lightpath is defined as the number of hops on the route ($CP = h$), and the wavelength assignment is the same as in case (a). Inversely, if there is no available wavelength on the route without using wavelength conversion, but there are available wavelengths on the segments and there is a combination of these segments which can establish a lightpath on the route by using wavelength conversion, then the cost of the primary lightpath is defined as the number of hops on the route plus $t_{wc} \cdot C_{wc}$ (i.e., $CP = h + t_{wc} \cdot C_{wc}$), where C_{wc} is the cost of a wavelength converter, and t_{wc} is the number of times wavelength converters are used on the route. Wavelengths of the combination will be assigned to the primary lightpath. Otherwise, CP is infinite.

When a route having wavelength converters serves as a backup route, the cost of the backup lightpath is computed as follows: Let $w_{(i,j)}$ be the j th wavelength on the i th segment of the route. $c_{f,w_{(i,j)}}^i$ denote the λ -channel of the wavelength $w_{(i,j)}$ on fibre f which belongs to i th segment. Its value is defined the same as the value of $c_{f,w}$ in Section 3.3.1. Then, the cost of the i th segment for each wavelength $w_{(i,j)}$, that is $C_{w_{(i,j)}}$, is computed as the sum of the values of the λ -channels of the route.

$$CB_{w_{(i,j)}}^i = \sum_{f \in \text{segment } i} c_{f,w_{(i,j)}}^i. \quad (6)$$

The cost of the backup lightpath is then defined as minimum of the total costs, summing up over the possible combinations of these segments of the route under the wavelength conversion constraint.

$$CB = \min \left\{ \sum_{i \leq t_{wc}+1; j=0,1,\dots,W; \text{abs}(w_{(i-1,j)}-w_{(i,j)}) \leq r} CB_{w_{(i,j)}}^i \right\} \quad (7)$$

where r is the number of conversion units of the wavelength converter.

According to the Eq. (7), the backup lightpath is straightforwardly traced according to the Viterbi algorithm [17].

The pseudo codes of functions to compute the cost of a primary paths and backup paths are as follows:

{Primary paths}

If (there is an available wavelength on the route) **Then**

$CP \leftarrow h$;

Else

If (there are available wavelengths on the segments and there is a combination of these segments which establishes a lightpath on the route) **Then**

$CP = h + t_{wc} \cdot C_{wc}$

Else

$CP = \infty$;

End if

End if

{Backup paths}

If (there is no a wavelength converter on the route)

Then

For each (wavelength w) **In** W

Compute CB_w according to (1)

EndFor

Compute CB according to (2);

Else

For each (segment i) **In** the route

For each (wavelength w) **In** W

Compute $CB_{w(i,j)}^i$ according to (6);

EndFor

EndFor

Compute CB according to (7);

EndIf

Similarly as in Section 3.3.1, a cycle ($s - d - s$) has two interpretations; the cost of the cycle is also computed assuming both interpretations and that with the lower cost is chosen. For each interpretation, with keeping in mind that the cycle that has the shortest available primary path is always chosen, the cost of a cycle is also defined as follows:

$$C = CP + \alpha \cdot CB. \quad (8)$$

Note that without wavelength conversion, t_{wc} is equal to zero and the computation of CP and CB is the same as in case (a) above. In other words, in terms of the cost function C , optical networks without wavelength conversion become a special case of optical networks with sparse wavelength conversion.

The value of C_{wc} should be chosen to ensure that a cycle having a primary route with a larger number of wavelength converters always gets a larger cost. Suppose that the primary route of one cycle has t_{wc} converters and its cost is CP_1 ; the primary route of another cycle has $(t_{wc} + 1)$ converters and its

cost is CP_2 , then the C_{wc} should meet the following requirement:

$$h_1 + t_{wc} \cdot C_{wc} < h_2 + (t_{wc} + 1) \cdot C_{wc}.$$

In the above requirement the cost of the backup route does not need to consider because $\alpha \cdot CB < 1$ is always required, and the parameter α is chosen as in Section 3.3.1.

The route corresponding to the primary lightpath that costs CP_2 has to accommodate at least one converter ($t_{wc} + 1 \geq 1$), thus the minimum of its number of hops is two. Nevertheless, the maximum number of hops on the route corresponding to the primary lightpath that costs CP_1 is L , where L is the length of the longest path of the network and the value L is determined off-line based on the algorithm described in [15]. Thus:

$$L < 2 + C_{wc}.$$

Consequently, the lower bound for C_{wc} is:

$$C_{wc} > L - 2. \quad (9)$$

3.4. Complexity of the hybrid algorithm

The value of parameter α is determined off-line before running the hybrid algorithm. The time complexity of the hybrid algorithm is dominated by its GA part.

Let P_{size} , G be the size of population and number of generations of GAs, N the number of network nodes, W the number of wavelengths on each link, and $N_c > 0$ the total number of converters in a network. The complexity of our hybrid algorithm can be estimated as follows:

The complexity of examining all possible pairs in the crossover stage is $P_{size}(P_{size} - 1)/2$ times the complexity of examining a pair of routes. This operation includes the following steps: search common nodes, create children, check if the children are valid cycles and different from their parents. All these steps require $O(N)$ time. Moreover, the complexity of evaluating the cost of a cycle is performed in two steps: (1) compute the cost of the primary lightpath whose complexity is $O(WNN_c)$ when there is wavelength conversion on the network, otherwise whose complexity is $O(WN)$ and (2) compute the cost of the backup lightpath. Since all wavelengths in all channels used by the backup lightpath are examined, and each channel may be shared by another backup lightpath, the result is that all links of the corresponding primary lightpath must be examined to check if there is a conflict. These operators require $O(WN^2)$ time units. Thus in the presence of wavelength converters the complexity of evaluating the

cost of the backup lightpath according to the Viterbi algorithm is $O(WN^2N_cr)$, where $r > 0$ is the number of conversion units of a wavelength converter. Therefore, the complexity of the crossover stage when there are wavelength converters on the network is $O((P_{\text{size}}(P_{\text{size}}-1)/2)N + P_{\text{size}}(WNN_c + WN^2N_cr)) = O(P_{\text{size}}^2N + P_{\text{size}}WN^2N_cr)$, otherwise the complexity of the crossover stage is $O((P_{\text{size}}(P_{\text{size}}-1)/2)N + P_{\text{size}}(WN + WN^2)) = O(P_{\text{size}}^2N + P_{\text{size}}WN^2)$.

In the mutation stage, a maximum of $P_{\text{size}} - 1$ routes are mutated. Thus, the complexity of the mutation stage with wavelength converters is $O(P_{\text{size}}(WNN_c + WN^2N_cr)) = O(P_{\text{size}}N^2N_cr)$, otherwise the complexity of the mutation stage is $O(P_{\text{size}}(WN + WN^2)) = O(P_{\text{size}}WN^2)$.

The reproduction stage only involves sorting the population by decreasing order of fitness values, so the complexity of this operation is $O(P_{\text{size}} \log P_{\text{size}})$.

In summary, the complexity of the hybrid algorithm in the presence of wavelength converters (i.e. $N_c > 0$ and $r > 0$) is:

$$O(G((P_{\text{size}}^2N + P_{\text{size}}N^2N_cr) + P_{\text{size}}WN^2N_cr + P_{\text{size}} \log P_{\text{size}})) = O(GP_{\text{size}}N(P_{\text{size}} + WNN_cr)).$$

Otherwise the complexity of the hybrid algorithm is

$$O(G((P_{\text{size}}^2N + P_{\text{size}}N^2) + P_{\text{size}}WN^2 + P_{\text{size}} \log P_{\text{size}})) = O(GP_{\text{size}}N(P_{\text{size}} + WN)).$$

4. Numerical results and analysis

In this section, we examine the performance of our new hybrid algorithm with an extensive simulation study based upon the *ns-2* network simulator [18] and the two typical network topologies illustrated in Fig. 6. With the presence of wavelength converters, we considered configurations that adopt two or five wavelength converters in each network topology. For each converter, we simulated the case of full-range wavelength conversion and also the cases of using one or two wavelength conversion units; here, a node with r conversion units can convert a wavelength w into any wavelength belonging to the set $[w - r, w + r]$.

To place the wavelength converters properly in a network, we used the placement scheme proposed in [16] to achieve the best benefit from the use of converters. In the case with two converters, the converters were placed in nodes three and five for the NSF network, and nodes zero and six for the EON network. For the case with five converters, the converters were placed in nodes 1, 3, 5, 10 and 12 for

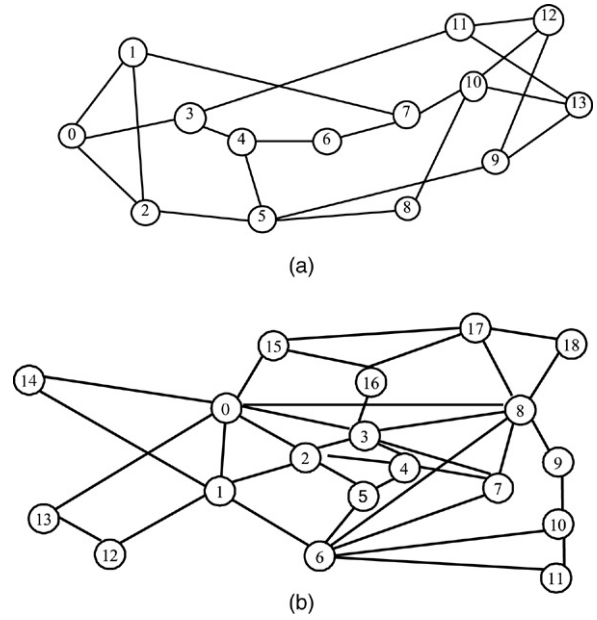


Fig. 6. Network topologies: (a) NSF network with 14 nodes and 21 links. (b) EON network with 19 nodes and 35 links.

the NSF network, and nodes 0, 1, 3, 6 and 8 for the EON network.

In our experiments, we used a dynamic traffic model in which connection requests arrive at a node according to a Poisson process with an arrival rate λ (call/seconds). The destination node was chosen randomly from all nodes except the source node. The connection holding time was exponentially distributed with mean holding time $1/\mu$ (s). The connection requests were distributed randomly on all the network nodes. Assume that each request requires a full wavelength bandwidth capacity.

There are N_s sessions over the network, so the total traffic load is $N_s * \lambda/\mu$ (Erlangs). Thus, we can modify the N_s , λ , and μ parameters to control workloads. In our simulations $N_s = 100$, the mean connection holding time is 60 s, that is, $1/\mu$. The average arrival rate, λ , is set depending upon the desired load. We also assume that all of links in the networks are bi-directional and each link has 8 wavelength channels ($W = 8$). We set $\alpha = 0.05$ to meet constraint (5) (since in NSF network we have $L = 12$, and $L = 18$ for EON network) and $C_{wc} = 17$ to meet the constraint (9). For the GA parameters G and P_{size} , we use the same values as in [7], i.e. we set $P_{\text{size}} = 8$ and $G = 8$, both for the FT-GRWA and our hybrid algorithm. For the mobile agents part of our hybrid algorithm, we simply adopt the same set of parameters as in [14] for pheromone table updating. Particularly, $\theta = 0.8$, $\beta = 1.75$, and $\gamma = 0.2$. The parameter t_{ant} is set depending upon the desired number

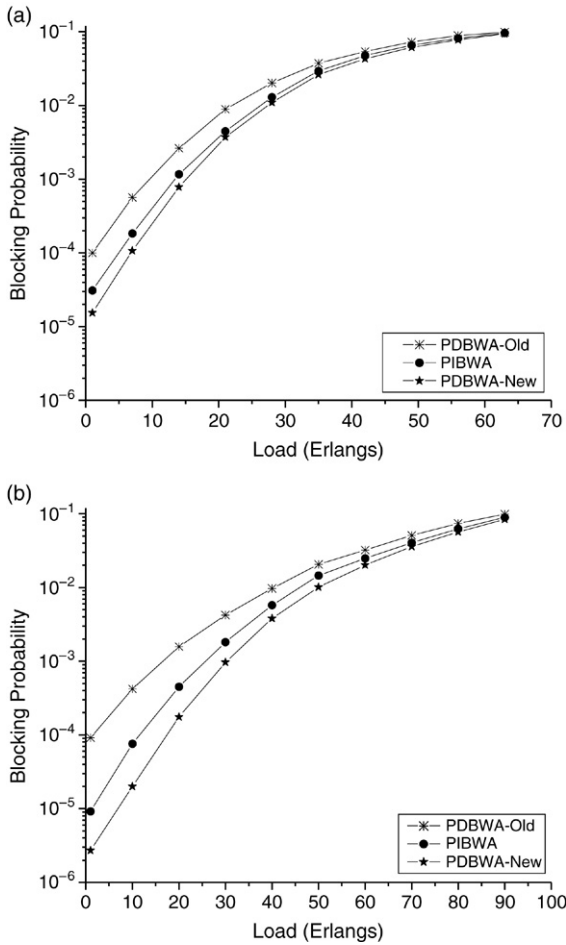


Fig. 7. Blocking probability vs. load in optical network without wavelength conversion. (a) NSF network. (b) EON network.

of generating agents. All the results, which we show below, are obtained with a 95% confidence level and a 4.53% confidence interval.

We first report the performance of the network in terms of blocking probability with using the PDBWA-old (i.e. original PDBWA algorithm), the PIBWA, and the PDBWA-New algorithms as shown in Fig. 7. In our simulations, all of these algorithms use three pre-computed alternate routes ($k = 3$). The comparison depicted in Fig. 7 clearly shows that the blocking probability of the PDBWA-New algorithm is significantly lower than that of the PIBWA or the original one for all cases. Thus, we only use the PDBWA-New for comparison with our hybrid algorithm.

It is necessary to adjust the number of mobile agents exploring the network states. The results of simulations about influencing the number of mobile agents generated at each node per second on the

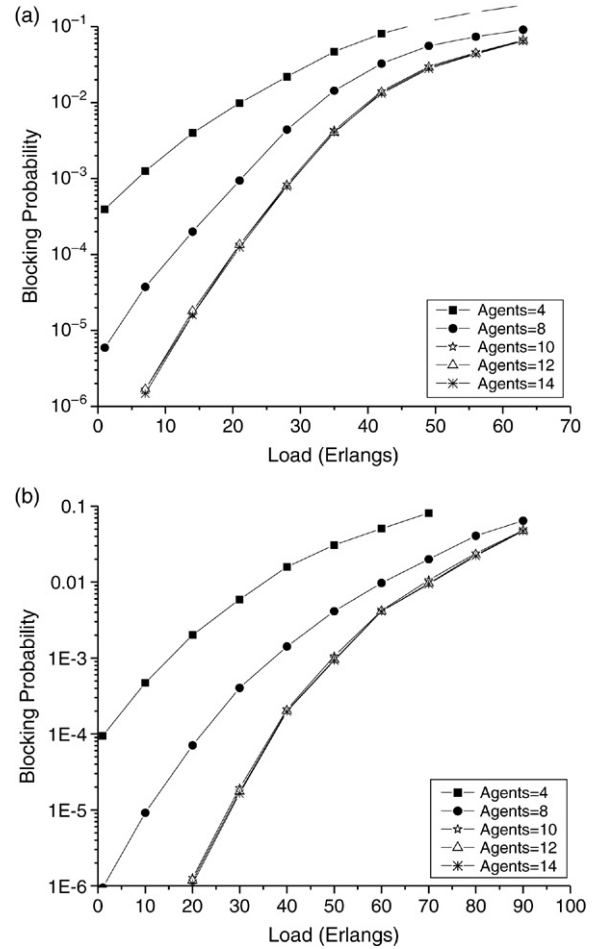


Fig. 8. Blocking probability vs. load in optical network without wavelength conversion. (a) NSF network. (b) EON network.

performance of the hybrid algorithm in terms of blocking probability are reported in Fig. 8.

We observe that when the number of generating agents is larger than 10, the blocking probability of the networks becomes stable. In all other simulations, we will keep the rate of generating agents as 10 per second at each network node.

To illustrate the influence of mobile agents on the performance of the network in terms of blocking probability, we report simulation results of the FT-GRWA algorithm using random search for generating the first population and compare with that using mobile agent as depicted in Fig. 9. As we have seen in Fig. 9, it is shown that using mobile agents to generate the first population did not increase the performance of the networks in terms of blocking probability, in comparison with using random search.

To illustrate the overall performance of our hybrid algorithm, Figs. 10–14 depict the comparison of the

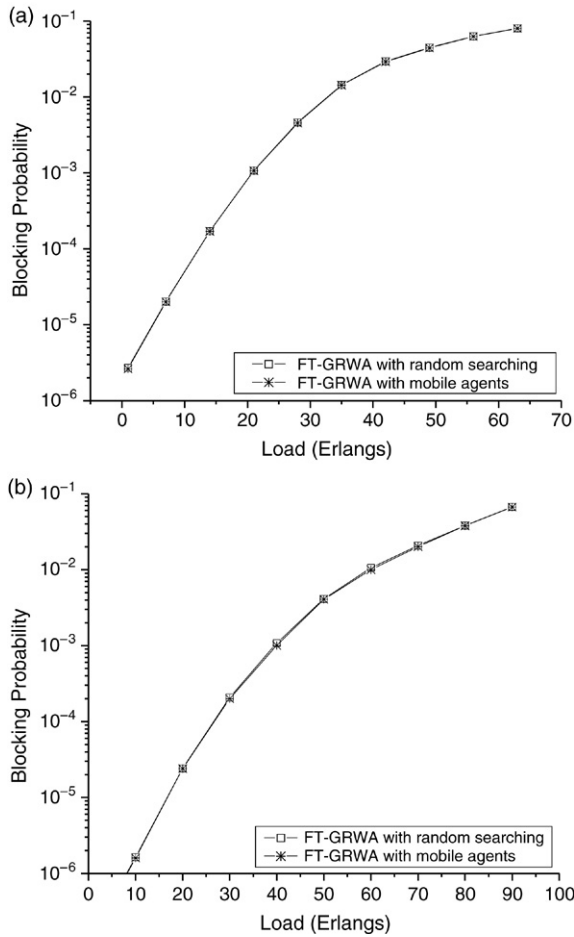


Fig. 9. Blocking probability vs. load in optical network without wavelength conversion. (a) NSF network. (b) EON network.

blocking probability between our hybrid algorithm, the PDBWA-New and the FT-GRWA algorithm. The results shown here may be different from those shown in [7] because the simulation scenarios are different. In this work, a request asks for the establishment of a bidirectional lightpath while in [7] a request asks for the establishment of a unidirectional lightpath.

For networks without wavelength conversion, the comparison depicted in Fig. 10 clearly shows that the blocking probability of the hybrid algorithm is significantly lower than that of the PDBWA-New or the FT-GRWA algorithm for all cases. For example, with the NSF network, the blocking probability of PDBWA-New and FT-GRWA algorithm is about 0.074 and 0.066 respectively, while that of our algorithm is only 0.038 when the workload is 56 Erlangs. Similar behaviours can be observed in the EON network. For instance, the blocking probability of the PDBWA-New and FT-GRWA algorithms is about 0.060 and 0.046 respectively

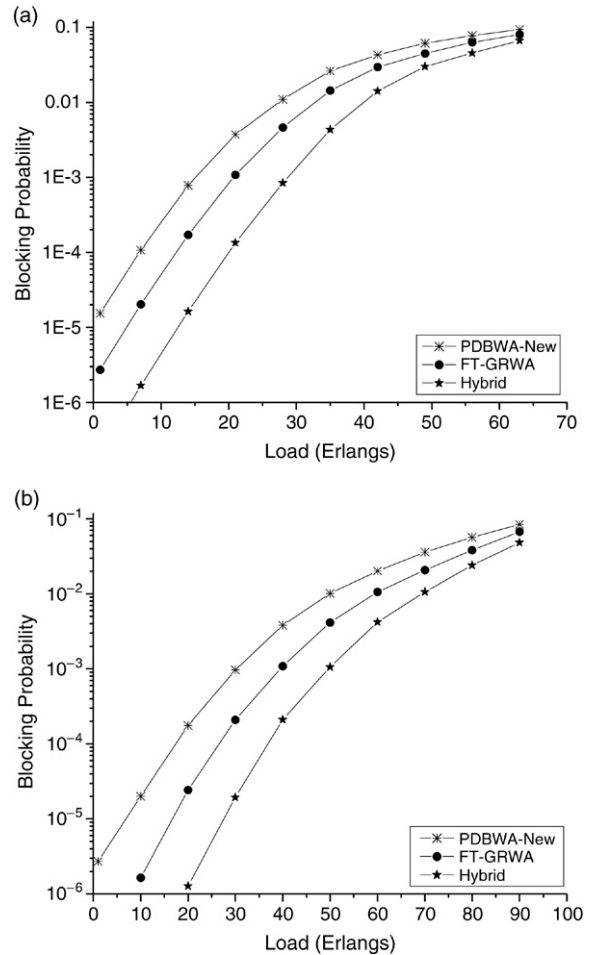


Fig. 10. Blocking probability vs. load in optical network without wavelength conversion. (a) NSF network. (b) EON network.

when the workload is 80 Erlangs, about twice higher than the blocking probability of our algorithm, which is 0.024.

Even with the gain from wavelength conversion, the simulation results in Figs. 11–14 show that the hybrid algorithm still achieves a lower blocking probability than the PDBWA-New or the FT-GRWA algorithm. Here, the original PDBWA-New and FT-GRWA algorithms did not consider the presence of wavelength conversion, thus in this work we modified those algorithms to use wavelength converters according to the same ideas as in our hybrid algorithm.

The comparison given in Figs. 11–14 shows that blocking probability with our hybrid algorithm is significantly lower for the cases of adopting two and five converters. We can also see that the improvement from using our algorithm decreases as the number of wavelength converters increases. For example, for the NSF network with two wavelength converters

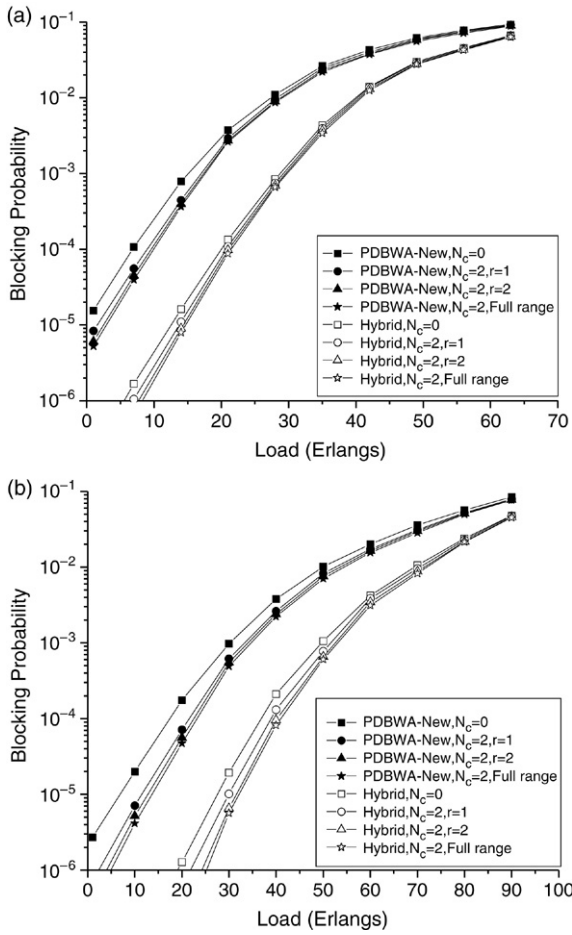


Fig. 11. The performance of the PDBWA-New algorithm in comparison with the hybrid algorithm on the networks having 2 wavelength converters. (a) NSF network. (b) EON network.

(Figs. 11(a) and 12(a)) and $r = 2$, the blocking probability of the PDBWA-New algorithm is about 0.062 and that of FT-GRWA algorithm is about 0.054, while that of our algorithm is only 0.031 with a workload of 56. For the same set of r and workload in the NSF network with five wavelength converters (Figs. 13(a) and 14(a)), the blocking probability is about 0.049 and 0.045 for the PDBWA-New and FT-GRWA algorithms respectively, and 0.027 for our algorithm. Similar behaviours can be observed in the EON network (Figs. 11(b), 12(b), 13(b) and 14(b)).

Figs. 11–14 also show that for all three algorithms, while we can always decrease blocking probability by adopting wavelength converters, this decrease is inversely proportional to the number of wavelength converters and the wavelength conversion capability of each converter. Actually, by adopting only limited-range wavelength converters, those algorithms can achieve performances similar to using full-range wavelength

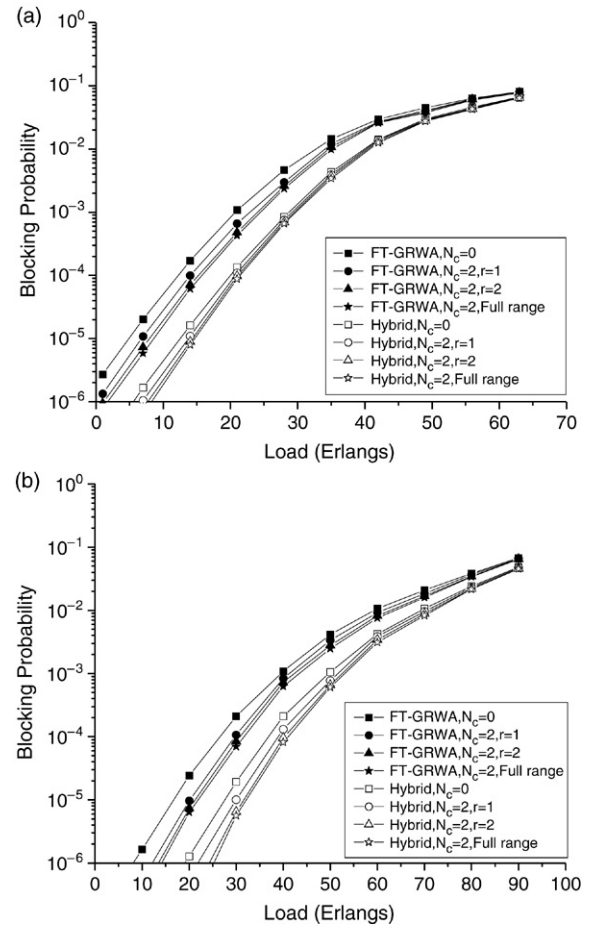


Fig. 12. The performance of the FT-GRWA algorithm in comparison with the Hybrid algorithm on the network having two wavelength converters. (a) NSF network. (b) EON network.

converters. For example, for the NSF network with two converters and a workload of 56, the blocking probability of the hybrid algorithm is 0.031 when $r = 2$, and this blocking probability is slightly reduced to 0.030 when we adopt full-range wavelength converters. For the EON network with five converters and a workload of 80, the blocking probability of the hybrid algorithm is 0.015 when $r = 2$, and we can only reduce this blocking probability slightly to 0.014 when we use full-range wavelength converters in the network.

The results in both Figs. 11 and 13 indicate that our new algorithm is a significant improvement over the PDBWA-New in terms of blocking probability. This is partially because the PDBWA-New algorithm used in our comparison has only three alternative cycles determined statically in advance, so it has very limited adaptability to the state variation of a dynamic network. On the other hand, each generation of our hybrid algorithm has eight feasible alternative cycles

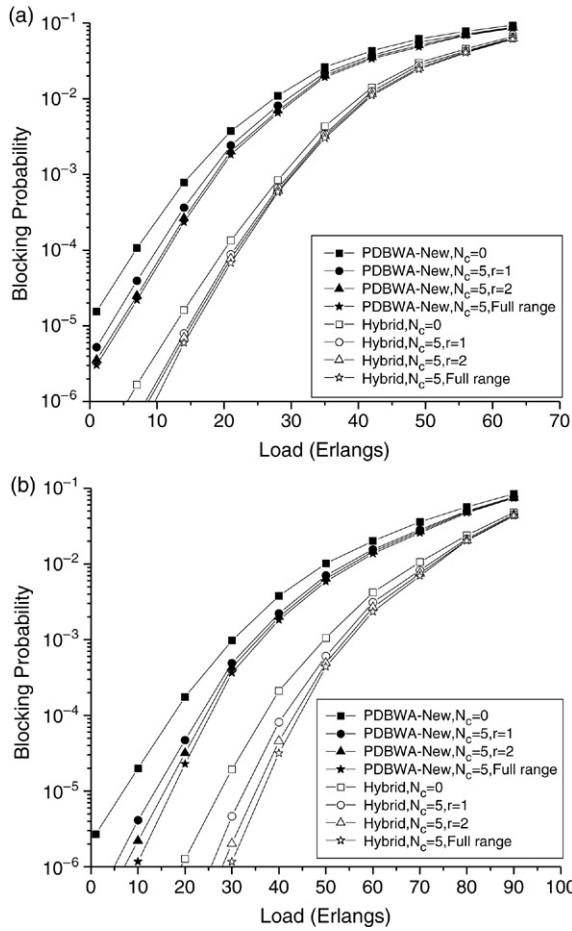


Fig. 13. The performance of the PDBWA-New algorithm in comparison with the hybrid algorithm on the network having five wavelength converters. (a) NSF network. (b) EON network.

($P_{size} = 8$) determined dynamically based on the current network state, so our algorithm has a much better adaptability to the state variation of a dynamic network and thus has more chances to find a cycle for a connection request. Another reason why the proposed algorithm works well in the numerical simulation is the powerful evolution capability of the GA combined with the new fitness function, which in turn further reduces the blocking probability as indicated in Figs. 10 and 12.

To show that our new algorithm can result in a significantly lower setup delay than the FT-GRWA algorithm, we summarize in Tables 1 and 2 the average execution time of a request for the NSF network and the EON network based on a simulation in a computer with a Pentium III 600 MHz processor, HDD 10 GB IDE and 512 MB RAM. For the sake of comparison, we also include in Tables 1 and 2 the average execution time of the PDBWA-New algorithm.

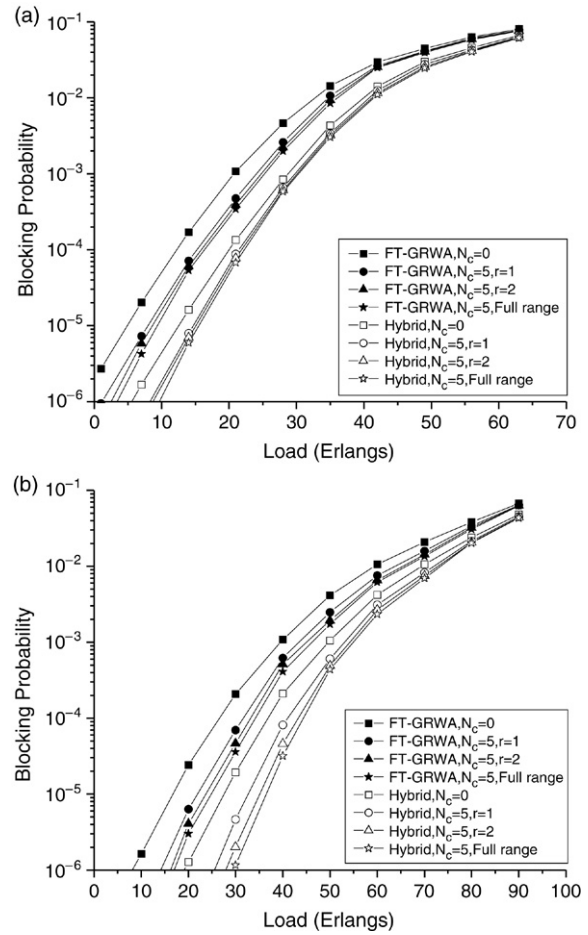


Fig. 14. The performance of the FT-GRWA algorithm in comparison with the Hybrid algorithm on the network having five wavelength converters. (a) NSF network. (b) EON network.

Note that, the results shown here are higher than the results described in [7], because we used the ns-2 simulator for our simulations while the authors in [7] used their own home-made simulator.

Here, because mobile agents run parallel and independently with GA to find cycles and store them in the cycle routing tables, we do not need to care about the time of finding a cycle by mobile agents.

The results given in Tables 1 and 2 indicate that for both the NSF and EON networks, the average execution time of our new algorithm is significantly lower than that of the FT-GRWA algorithm. For the NSF network with a workload of 56 Erlangs, the average execution time of the FT-GRWA algorithm is 20.57 ms, which is significantly higher than the 12.15 ms average execution time of the hybrid algorithm. When the EON network works under a workload of 80 Erlangs, the average execution time of the FT-GRWA algorithm is 19.02 ms,

Table 1

Average execution time of dynamic survivable RWA algorithms in the NSF network (ms)

N_c	r	A	Load (Erlangs)				
			35	42	49	56	63
			Time (ms)				
0		F	17.67	17.81	19.39	20.57	22.88
		H	10.80	11.18	11.98	12.15	13.77
		P	1.54	1.60	1.71	1.74	1.97
2	1	F	27.79	28.09	30.70	31.57	35.23
		H	17.51	17.89	19.17	20.01	22.81
		P	2.44	2.54	2.68	2.73	3.05
	2	F	29.78	30.09	32.89	33.82	37.74
		H	18.76	19.17	20.54	21.44	24.43
		P	2.62	2.72	2.92	2.87	3.27
	Full range	F	31.76	32.10	35.08	36.07	40.26
		H	20.01	20.45	21.91	22.86	26.06
		P	2.79	2.90	3.06	3.12	3.48
5	1	F	31.50	31.52	34.54	34.55	38.82
		H	20.65	21.32	22.27	22.38	26.47
		P	2.58	2.66	2.74	2.86	3.14
	2	F	33.59	33.62	36.84	36.85	41.41
		H	22.02	22.74	23.75	23.87	28.23
		P	2.77	2.84	2.93	3.07	3.36
	Full range	F	35.69	35.72	39.14	39.16	44.00
		H	23.40	24.16	25.23	25.36	29.99
		P	2.95	3.03	3.13	3.27	3.58

Table 2

Average execution time of dynamic survivable RWA algorithms in the EON network (ms)

N_c	r	A	Load (Erlangs)				
			50	60	70	80	90
			Time (ms)				
0		F	16.42	17.26	18.83	19.02	20.44
		H	9.36	10.45	10.62	11.22	11.25
		P	1.82	1.92	2.09	2.11	2.27
2	1	F	23.39	25.32	28.02	27.71	30.07
		H	15.51	17.38	18.14	19.34	19.42
		P	2.62	2.83	3.10	3.13	3.36
	2	F	25.06	27.13	30.02	29.69	32.22
		H	16.54	18.54	19.35	20.63	20.71
		P	2.80	3.03	3.32	3.35	3.60
	Full range	F	26.73	28.93	32.02	31.67	34.37
		H	17.57	19.69	20.56	21.92	22.00
		P	2.99	3.23	3.54	3.58	3.84
5	1	F	23.86	26.04	28.75	28.12	30.61
		H	16.06	18.17	19.11	20.22	20.41
		P	2.67	2.91	3.14	3.21	3.42
	2	F	25.56	27.90	30.81	30.12	32.79
		H	17.13	19.38	20.38	21.57	21.77
		P	2.86	3.12	3.36	3.44	3.66
	Full range	F	27.27	29.76	32.86	32.13	34.98
		H	18.20	20.59	21.65	22.92	23.13
		P	3.05	3.32	3.59	3.67	3.90

which is also significantly higher than the average execution time of the new algorithm (11.22 ms).

It is notable in the two tables above that the PDBWA-New algorithm requires a much smaller execution time than either the FT-GRWA algorithm or our hybrid algorithm, because the PDBWA-New algorithm used in our comparison has only three alternative cycles determined statically in advance and it does not involve any evolution process to find the best cycle for a request. However, the smaller execution time of PDBWA-New comes at the price of a significantly higher blocking probability than the new hybrid algorithm, as indicated in Figs. 10, 11 and 13. Though the execution time of the hybrid algorithm is ten times more than that of the PDBWA-New algorithm, it should be recommended to use the proposed hybrid algorithm because the execution time for routing is quite small in comparison with the total time needed to set up a connection in the real world, which includes the execution time of the algorithm, the delay time, the processing time, and the queuing delay. Moreover, the bandwidth in WDM networks is huge (few gigabits per second); thus priority for efficient resource utilization is more important than priority for the execution time of an algorithm for each request. It would also be worth noting that the execution time of algorithms will be significantly lower than that given in Tables 1 and 2 in a real network, because the algorithm will not be implemented in the ns-2 environment.

In addition, using the mobile agents technique not only reduces the time consumed by generating the first population through the random searching associated with available GA-based routing algorithms, but also decreases the time of the evolution process of those algorithms, as described in Fig. 15.

Fig. 15 shows that for the NSF network, the probability of fulfilling the stopping criterion at the first iteration of the hybrid algorithm is 91.03%, which is larger than the probability of fulfilling the stopping criterion at the first iteration of the algorithm used for random searching in generating the first population, 78.68%. Similarly, for the EON network, the two values are 93.76% and 82.61% respectively. This shows that with the hybrid algorithm, the good first population results in a smaller probability that the evolution process will occur, compared to FT-GRWA, and this is another reason why we used the mobile agents technique in our hybrid algorithm.

5. Conclusions

In this paper, we have proposed a novel hybrid algorithm for dynamic survivable routing in WDM

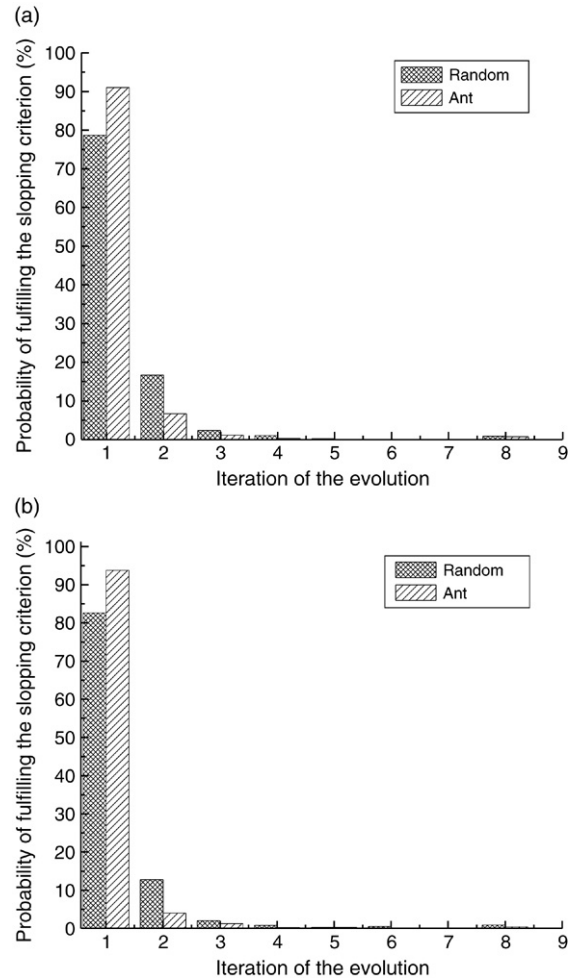


Fig. 15. Probability of fulfilling the stopping criterion in each iteration of the evolution process. (a) NSF network with traffic load of 35. (b) EON network with traffic load of 60.

optical networks. By combining the mobile agents technique with an appropriate genetic algorithm, we have produced a hybrid algorithm that can reduce the time consuming process of generating the first population of cycles as well as reduce the probability that the evolution process has to occur while retaining the GA's attractive ability to achieve a significantly low blocking probability. We also proposed a new fitness function for the genetic algorithm. With this fitness function, the hybrid algorithm is able to find a cycle for dependable lightpath setup so that network resources are used efficiently, thus significantly improving network performance compared with other algorithms. Moreover, the general formula for determining the key parameter in the new fitness function also is proposed. Extensive simulation results showed that the new hybrid algorithm can achieve a lower blocking probability than

either the PDBWA-New or FT-GRWA algorithm on NSF and EON networks, both without and with sparse wavelength conversion. The simulation results also indicated that the setup time of a request can be reduced with our hybrid algorithm, compared to FT-GRWA.

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